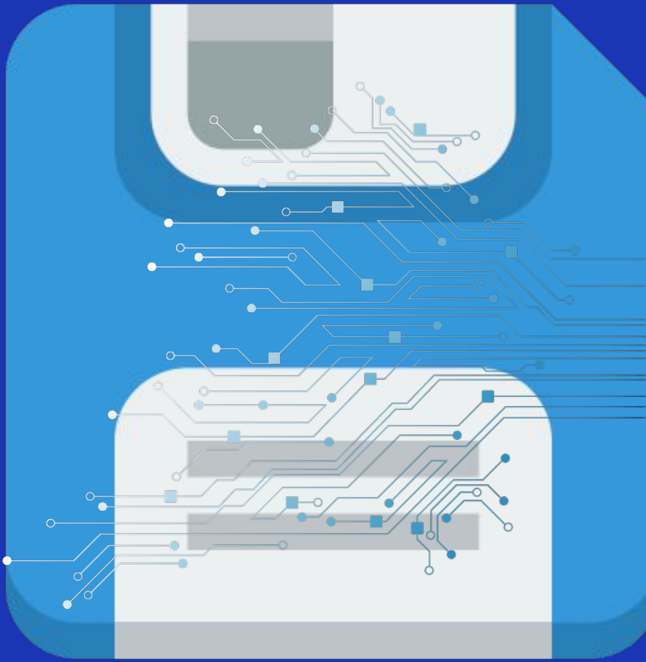
The SAVURe

Let the SAVURe be you Savior!

Self-Contained
Autonomous
Version-
Updating
Robot



Nathan Janssen: MAE 6291 Midterm Project

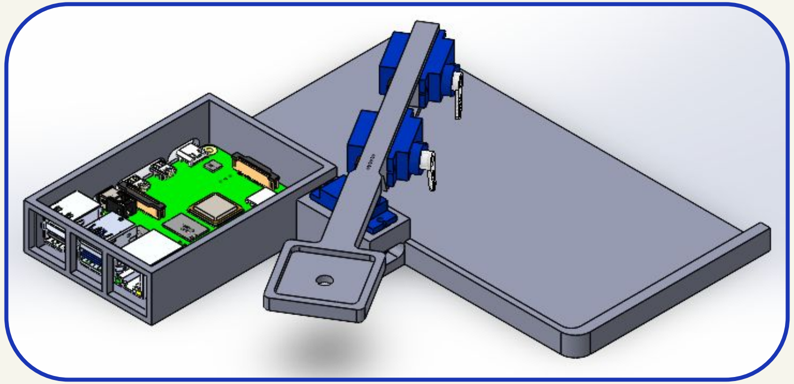
The “Thing”

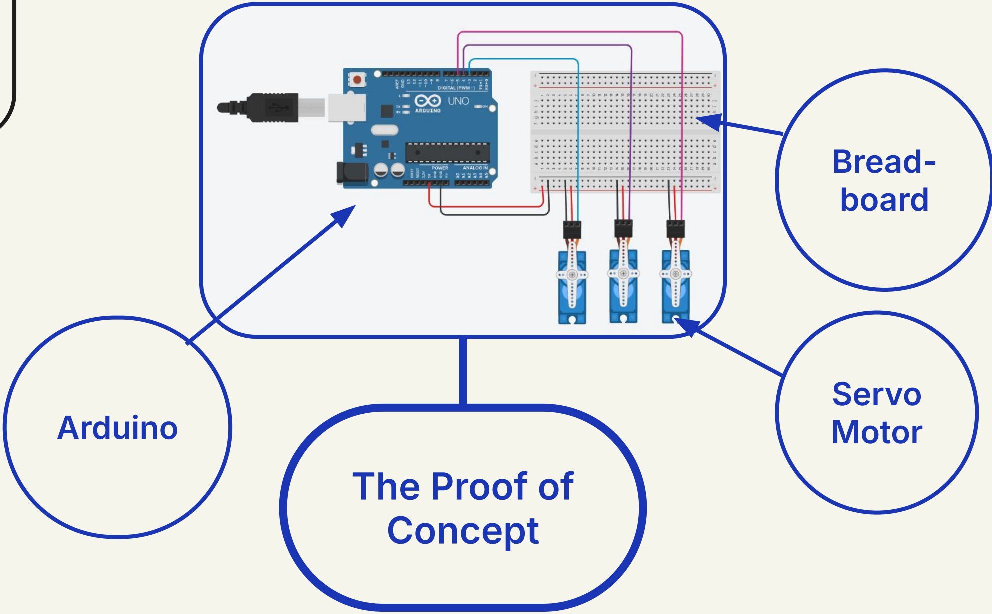
The Design:

- Saves user projects remotely
- A 3D printed base to interface with the laptop
- A case with cutouts to ensure the RPi is secure
- 3D Printed arm to house servos
- 3 Micro Servos to produce the movement

Hardware Used:

- Raspberry Pi 4 Model B
- Breadboard
- 3 Servo Motors





Software:

Programming the Functions:

```python
def move_servo_slowly(servo, start_angle, end_angle, step_delay=0.05, step_size=2):
    """ Moves the servo from start_angle to end_angle slowly in small steps """
    if start_angle < end_angle:
        step = step_size
    else:
        step = -step_size

    for angle in range(start_angle, end_angle + step, step):
        duty = angle_to_duty_cycle(angle)
        servo.ChangeDutyCycle(duty)
        time.sleep(step_delay)


    servo.ChangeDutyCycle(angle_to_duty_cycle(end_angle))
    time.sleep(0.5)
```

+

```python
def set_angle(servo, angle):
    duty = 2.5 + (angle / 18.0)
    servo.ChangeDutyCycle(duty)
    time.sleep(1)
```

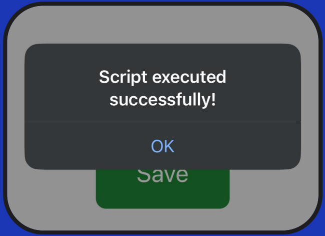
Consists of:

1. Slow controlled motion
2. Quick angle changes

Creating a User Interface:

```python
from flask import Flask, render_template
import os

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/save')
def save():
    os.system('python3 /home/pi/Desktop/spring2025_codes/midterm.py')
    return "Script executed successfully!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)
```

→



1. Run Flask to initiate app
2. Opens html app interface
3. The app is put online using ngrok

3-Layer IoT Model

Layers

The IoT 3-Layer Model chosen consists of actuation, network, and application. This is because no sensors are involved and therefore there is no sensor layer.

PERCEPTION/ ACTUATION

The servo motors interact with the physical world, demanding information from the network in order to run

NETWORK

The network consists of a Flask code run from the RPi to execute the code and a ngrok server to connect the app to the internet

APPLICATION

The Flask code connects to an html app that includes on save button and notifies the user when the script has been successfully run

Moving Forward

1st

Consistent

Aim to convert the url that the app needs in order to get online from changing every use to one constant url.

2nd

Run on Boot

Implement a code to run the app initiation and ngrok deployment of the url on boot.

3rd

Autonomous

Implement a code that activates the servo arm every 5 minutes during a specified time that a project is open.

4th

Touch Grass